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MIL-DTL-85097/11D(OS)
13 September 2000
Superceding
MIL-A-85097/11C(OS)
24 October 1994

MILITARY ASSOCIATED DETAIL SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR ASSEMBLY, MK 113 MODS

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Mk 113 Mods Rocket Motor Assembly described herein shall consist of this specification and the latest issue of MIL-DTL-85097.

1. SCOPE

1.1 Scope. This specification covers the requirements unique to the Rocket Motor Assembly Mk 113 Mods (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-DTL-85097 USN/USMC Propellant Actuated Devices (Pads), General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 840M), Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377
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(Unless otherwise indicated, copies of Department of Defense specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications are in addition to those listed in the basic specification, MIL-DTL-85097, and form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2)

SPECIFICATIONS

NAVAL SURFACE WARFARE CENTER (CAGE Code 14083)

511-174TD-004 Propellant, NOSIH-BC-11, NSWC/IH-EC-02, NSWC/IH-EC-04 and NSWC/IH-EC-05

510-174TD-004 Propellant, NSWC/IH-EC-8

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL673AS200	Rocket Motor Assembly, Mk 113 Mods
DL673AS205	WORD Rocket Motor Assembly
673AS200	Rocket Motor Assembly, Mk 113 Mods
673AS205	WORD Rocket Motor Assembly

(Application for copies of specifications and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

OP 2165 Navy Transportation Safety Handbook for Hazardous Materials

(Application for copies should be addressed to the Aviation Supply Office, Naval Publications and Forms Directorate, Attn: 1053, Building 26, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements in this associated detail specification, unless in direct conflict with the basic specification, MIL-DTL-85097, shall be considered additional, not substitute. The Rocket Motor

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Assembly Mk 113 Mods shall conform to DL673AS200 and the Mk 113 Wind Oriented Rocket Deployment (WORD) Motor Assembly shall conform to DL673AS205. The Mk 113 Mod 0 shall contain NOSIH-BC-11 propellant. The Mk 113 Mod 1 shall contain NSWC/IH-EC-8 propellant. If there is a direct conflict with the basic specification, this document shall take precedence.

3.2 Item description. The Rocket Motor Assembly Mk 113 Mods specified herein is comprised of a WORD rocket motor mated to a drogue release mechanism.

3.3 Propellant. The propellant shall conform to 511-174TD-004 or 510-174TD-004. The contractor shall submit all test results in accordance with 511-174TD-004 or 510-174TD-004 and MIL-DTL-85097.

3.4 Propellant aging. When tested in accordance with MIL-DTL-85097, propellant shall meet the aging requirements specified in 511-174TD-004 for the Mk 113 Mod 1 or 510-174TD-004 for the Mk 113 Mod 0. Testing shall conform to paragraph 4.4.

3.5 WORD motor ballistic performance. The ballistic performance shall conform to Table I.

TABLE I. Ballistic performance requirements.

Characteristic	Value		Temperature (± 5°F)
	Min	Max	
Impulse over 390 ms (lb-s) (see 6.2.1)	47	--	-40, 160
Ignition delay (ms)	-	10	-40, 160
Actuation force (lb)	15	25	-40, 160
Rise time (ms) (see 6.2.2)	-	30	-40, 160
Thrust (lb)	-	270	-40, 160

3.6 Rocket motor assembly test/Drogue release mechanism test. Ten drogue release mechanisms shall be assembled onto an explosives vacated WORD rocket motor in order to test: (1) the WORD rocket motor assembly, (2) the drogue release mechanism. The government shall provide such motors. Performance shall be in accordance with paragraph 3.7. Testing shall be in accordance with paragraph 4.5. See test definitions in sections 6.4 and 6.5.

3.7 Drogue release mechanism performance . The drogue release test configuration shown in Figure 1 shall be employed. The drogue release mechanism shall actuate within 400 to 1,000 psig and shall in no way hinder deployment of the WORD rocket motor in any configuration/load arrangement.

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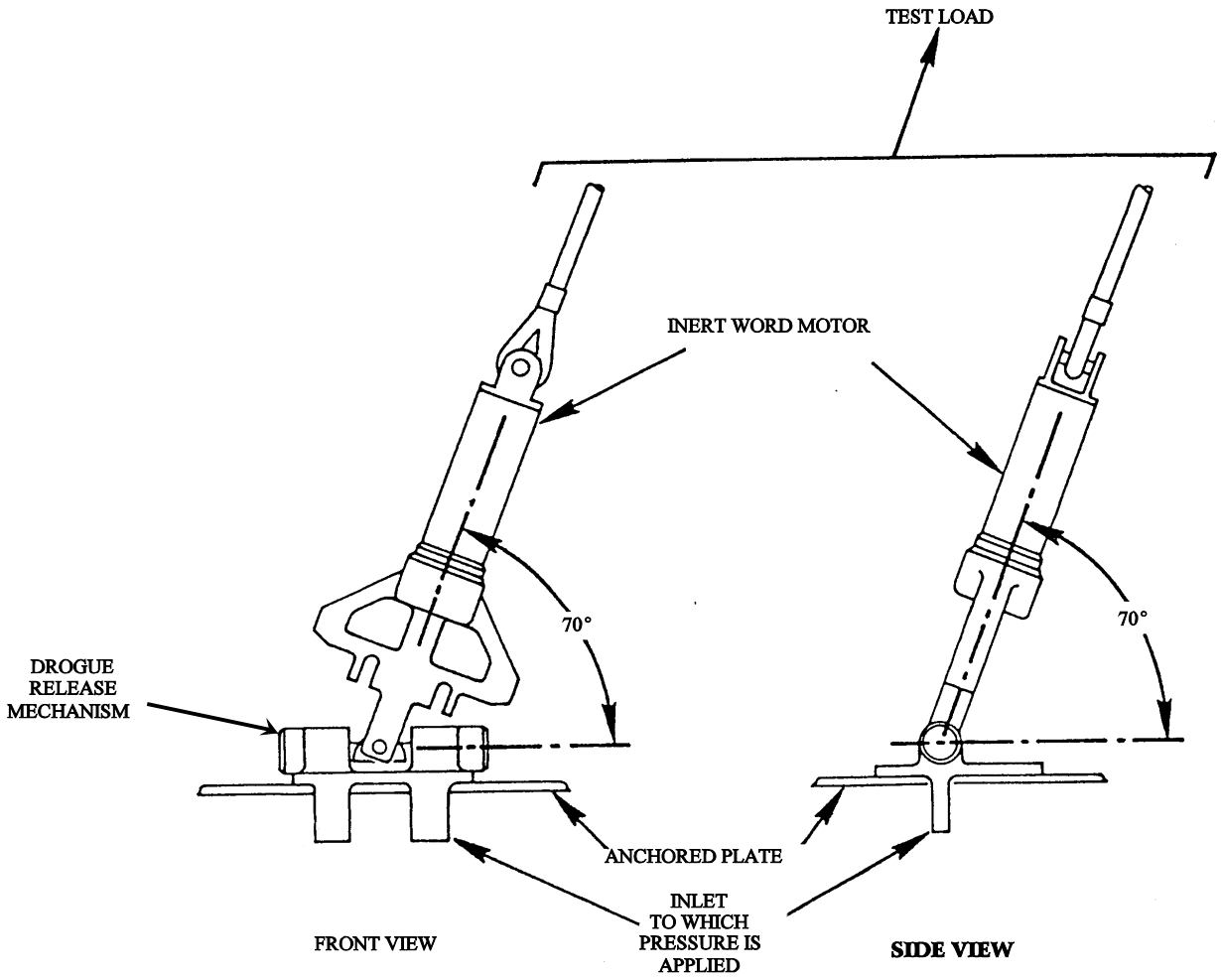


FIGURE 1. Drogue release test configuration.

4. VERIFICATION.

4.1 Unit testing. Unit testing includes the following phases:

1.) Propellant aging
2.) WORD motor ballistic performance
3.) Rocket motor assembly test/Drogue release mechanism test

4.2 Conformance inspection. The conformance inspection shall consist of the inspections and tests listed in 4.3, 4.4 and 4.5.

4.2.1 Conditioning period. Time for temperature conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: not less than 3 hours.
b. Temperature cycling: not less than 4 hours.

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4.3 Radiographic inspection. Units shall be radiographed in two views 90° apart, as specified by Drawing 673AS200 or by Drawing 673AS205, as appropriate.

4.4 Propellant aging. Propellant aging testing shall be in accordance with MIL-DTL-85097. Units shall meet the requirements of 3.4.

4.5 Drogue release mechanism testing and rocket motor assembly testing. Ten drogue release mechanisms mated to an explosives vacated WORD rocket motor shall be submitted for each test. Testing shall be performed on the drogue release mechanism and the rocket motor assembly in accordance with the following:

a. Condition 50% of the test units at -40°F and condition 50% at 160°F for the time specified in 4.2.1.

b. There shall be a light test load of 25 to 35 lb, and a heavy test load of 1000 to 1025 lb. The distribution of units tested with the light and heavy loads shall conform to Table II.

Table II. Distribution of light and heavy loads.

Quantity	Load (lb.)	Temperature (±5°F)
3	1000 - 1025	-40
2	25 - 35	-40
2	1000 - 1025	+165
3	25 - 35	+165

c. Actuation pressure shall be regulated to 1,000 to 1,025 psig and pressurization rate shall be regulated to 30,000 to 50,000 psig/sec.

d. Only one side port of each drogue release mechanism shall be tested. However, 50% of the test units shall be tested on the right port only and 50% of the test units shall be tested on the left side port only. These test units will be randomly chosen. The load arrangement in Figure 1 shall be utilized.

5. PACKAGING

(This section is not applicable to this specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Mk 113 Mods Rocket Motor Assembly assists in the deployment of the main parachute in T/AV-8B, A-7E, and TA-7C aircraft. The rocket motor assembly was developed for use in only those military aircraft, and therefore has no commercial application.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

c. The Mark and Mod designation of the rocket motor.

6.3 Definitions. For the purpose of this specification, the following definitions apply:

6.3.1 Impulse over 390 ms. The total impulse over the interval from the start of the burn time (time at 10% of the maximum thrust during the initial rise on the thrust curve t_{b0}) to 390 ms of burn time (t_{b0} + 390 ms) or the end of the burn time, whichever occurs first.

6.3.2 Rise time. The interval from 10% to 75% of the ignition peak thrust.

6.3.3 Explosives vacated rocket motor. A previously fired rocket motor that has had all explosives and explosive residue removed and has been cleaned. It differs from an inert in that it is not painted blue, and in order to retain its strength it does not have holes drilled through it.

6.4 Rocket motor assembly test. The test performed to ensure that the manufacturer of the WORD rocket motor assembly has properly assembled the drogue release mechanism onto the rocket motor during production. The manufacturer should intersperse the explosives vacated motors into the production assembly process.

6.5 Drogue release mechanism test. The test performed to ensure that the drogue release mechanisms are acceptable for use as future Government Furnished Material.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-85097/11 D	2. DOCUMENT DATE (YYMMDD) 13 September 2000

3. DOCUMENT TITLE
AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR ASSEMBLY, MK 113 MODS

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial: 301-744-4700 (2) DSN: 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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